Deep Reinforcement Learning-Based Physiological Control for Ventricular Assist Devices

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***Abstract*—With global population aging, the number of patients with end-stage heart failure continues to increase, and ventricular assist devices have become an important treatment option. However, most existing clinical devices still rely on constant pump speed control and cannot dynamically respond to changes in patient activity or resting state, nor can they adequately adapt to inter-patient physiological variability. These limitations reduce physiological perfusion matching and increase the risk of severe complications such as suction, regurgitation, and hemolysis. To address these challenges, this study presents a deep reinforcement learning-based physiological control method for ventricular assist devices. The proposed approach adopts an actor-critic framework with a hybrid CNN-Transformer neural network, where the CNN module extracts local features from physiological signals and the Transformer module captures long-range temporal dependencies. This architecture enables accurate perception of complex hemodynamic states and adaptive pump speed regulation. The proposed method is expected to improve the physiological adaptability, control safety, and hemodynamic performance of ventricular assist devices.**

***Index Terms*—ventricular assist device, physiological control, deep reinforcement learning, actor-critic, CNN-Transformer, hemodynamic regulation**

I. Introduction

Heart failure remains a major global health challenge, and the growing aging population continues to increase the number of patients with end-stage cardiac dysfunction [1]. Ventricular assist devices (VADs) have become an important therapy for these patients because they can provide mechanical circulatory support when the native heart is unable to maintain sufficient perfusion [3]. However, most currently deployed VADs still operate under constant pump speed settings. Although this strategy is simple and reliable, it cannot dynamically match the patient's changing physiological demand during transitions such as rest, exercise, and recovery [2], [3]. As a result, constant-speed control may lead to insufficient physiological perfusion and may increase the risk of adverse events including suction, regurgitation, and hemolysis.

To improve the physiological adaptability of VADs, many studies have explored closed-loop control strategies. Traditional methods, including pressure-based feedback control, fuzzy control, and Starling-like control, can improve performance under specific operating conditions [4]–[7]. Nevertheless, these approaches usually depend on simplified cardiovascular models, handcrafted rules, or manually tuned parameters. Because the cardiovascular system is highly nonlinear, time-varying, and patient-specific, model-based or rule-based controllers often struggle to maintain robust performance under complex hemodynamic disturbances. In addition, control objectives in VAD applications are inherently multi-objective, requiring the controller to simultaneously satisfy perfusion demand, maintain hemodynamic stability, and avoid complications. These requirements make conventional control design increasingly difficult.

Recent advances in artificial intelligence have created new opportunities for physiological control of biomedical devices [8], [9]. In particular, deep reinforcement learning (DRL) is attractive for VAD control because it can learn control policies directly from data without requiring an accurate explicit model of the cardiovascular system. Existing DRL-based studies have demonstrated the potential of reinforcement learning for pump speed regulation [2], [14], [16], but important limitations remain. Some methods rely on multilayer perceptrons or recurrent architectures with limited ability to jointly capture local waveform patterns and long-range temporal dependencies in physiological signals. Other methods focus on a narrow set of reward objectives and therefore do not fully reflect the safety-critical requirements of VAD operation. Consequently, there is still a need for a control framework that combines stronger physiological state representation with a more comprehensive reward design.

To address these issues, this paper proposes a deep reinforcement learning-based physiological control method for ventricular assist devices. The proposed approach adopts an actor-critic framework with a hybrid CNN-Transformer architecture, where the CNN module extracts local features from pressure and flow signals and the Transformer module models long-range temporal dependencies across cardiac cycles. In addition, a multi-objective reward function is designed to encourage adequate physiological perfusion while reducing the risks of suction, regurgitation, and hemolysis. The main contributions of this work are as follows. First, a CNN-Transformer actor-critic control architecture is introduced for VAD physiological control to improve hemodynamic state perception from sequential physiological signals. Second, a multi-objective reward design is developed to integrate physiological regulation and complication avoidance into a unified learning framework. Third, the proposed method provides a foundation for more adaptive, safe, and intelligent pump speed regulation in next-generation VAD systems.

II. Related Work

Existing VAD control strategies have evolved from constant-speed operation to more advanced physiological control schemes [3]. Conventional closed-loop methods, including pressure-based feedback control, proportional-integral regulation, fuzzy control, and Starling-like control, have shown the ability to improve pump response under specific operating conditions [4]–[7]. These methods are attractive because of their conceptual simplicity and clear control objectives. However, they typically depend on simplified cardiovascular models, manually selected control indicators, or extensive parameter tuning. As a result, their performance may degrade when confronted with the highly nonlinear, time-varying, and patient-specific nature of cardiovascular dynamics. In addition, traditional approaches often optimize a limited set of variables and therefore face difficulties when multiple safety and perfusion objectives must be satisfied simultaneously.

With the development of artificial intelligence, learning-based methods have been introduced to improve state perception and decision making for VAD regulation. Some studies have used recurrent neural networks to classify abnormal and normal physiological states, which helps enhance physiological signal interpretation but is still closer to state recognition than to full closed-loop control [8]. Other works have incorporated neural networks into existing control architectures to compensate for uncertainties or estimate hidden hemodynamic variables [10]. Convolutional neural networks have also been explored for sensorless estimation tasks, such as preload estimation from pump flow waveforms, demonstrating the potential of deep models to extract informative features from physiological signals [12]. Nevertheless, many of these approaches focus on diagnosis, estimation, or auxiliary modeling rather than directly learning an adaptive physiological control policy.

Deep reinforcement learning has further advanced the development of intelligent VAD controllers by offering a data-driven framework for continuous decision making. Recent studies based on PPO, DDPG, and TD3 have shown that reinforcement learning can regulate pump speed according to changing physiological conditions and can improve the adaptability of VAD operation [2], [14], [16]. Despite this progress, important limitations remain. In several existing studies, the actor and critic networks are built on multilayer perceptrons or recurrent modules that cannot simultaneously capture local waveform variations and long-range temporal dependencies with sufficient effectiveness [2], [14]. Moreover, some reward designs emphasize only a narrow subset of performance objectives, such as average flow or selected hemodynamic indicators, and therefore do not fully address the safety-critical requirements of VAD control, including the prevention of suction, regurgitation, and hemolysis.

Motivated by these gaps, this work aims to improve both physiological state representation and control objective design in deep reinforcement learning-based VAD regulation. The proposed method combines a CNN-Transformer architecture with an actor-critic framework to jointly model local signal patterns and long-term temporal information. In addition, a multi-objective reward function is introduced to better integrate physiological perfusion demand with safety constraints. In this sense, the present work extends existing research from general intelligent control toward a more comprehensive physiological control framework for ventricular assist devices.

III. Methods

This section presents the proposed deep reinforcement learning-based physiological control framework for ventricular assist devices. The overall method formulates VAD regulation as a continuous control problem and combines an actor-critic learning strategy with a CNN-Transformer state representation model and a multi-objective reward mechanism.

A. Problem Formulation

The VAD physiological control task is formulated as a sequential decision-making problem. At each control step, the controller receives a state vector composed of measured or estimated physiological signals, such as pressure- and flow-related waveforms and their recent temporal evolution. Based on this information, the controller outputs a continuous pump speed command to regulate the mechanical support provided by the device. The control objective is not only to maintain adequate perfusion under changing physiological demand, but also to preserve hemodynamic stability and reduce the risk of adverse events.

From a reinforcement learning perspective, the physiological state of the cardiovascular system defines the environment state, the pump speed adjustment corresponds to the action, and the immediate control quality is reflected by a reward signal. Because VAD regulation must continuously adapt to nonlinear and patient-specific dynamics, the control policy is required to learn from sequential interactions rather than from a fixed analytical model. This formulation allows the proposed method to address dynamic transitions such as rest-to-exercise changes and other time-varying physiological conditions in a unified framework.

B. Actor-Critic Framework

The proposed controller is built on an actor-critic reinforcement learning framework, which is well suited for continuous control problems such as pump speed regulation. The actor network maps the current physiological state to a control action, while the critic network evaluates the long-term utility of the policy by estimating the expected return under the current state-action relationship. Through iterative interaction with the environment, the actor is updated to generate more effective physiological control actions, and the critic provides guidance to stabilize policy learning. This design follows the general line of recent DRL-based VAD control studies while extending them with a stronger state representation module [2], [14], [16].

Compared with rule-based or purely supervised approaches, the actor-critic framework offers two main advantages in this problem. First, it can directly optimize sequential control performance under long-term physiological objectives instead of relying on manually designed control laws. Second, it can naturally support adaptive decision making in the presence of nonlinear cardiovascular dynamics and operating condition changes. This property is particularly important for VAD control, where the controller must continuously balance support effectiveness and safety under uncertain physiological disturbances.

C. CNN-Transformer Architecture

To improve physiological state representation, a hybrid CNN-Transformer architecture is adopted in the actor-critic framework. The CNN module is responsible for extracting local morphological features from physiological waveforms. This design is beneficial for identifying short-term signal variations and waveform patterns that may reflect instantaneous hemodynamic changes or early signs of unsafe operating conditions. In the VAD setting, such local information is important because adverse events may first appear as subtle disturbances in pressure or flow signals. The use of convolutional representations is also motivated by prior sensorless estimation studies that showed CNN-based models can effectively capture informative waveform features in VAD applications [12].

The Transformer module is introduced to capture long-range temporal dependencies in sequential physiological data. Unlike architectures that mainly focus on local or short-memory representations, the Transformer can model interactions across multiple cardiac cycles and therefore provide a more comprehensive description of evolving hemodynamic states. By combining CNN-based local feature extraction with Transformer-based temporal modeling, the proposed architecture is designed to improve physiological state perception and provide richer representations for both the actor and critic networks. This design is intended to address the representational limitations observed in previous DRL controllers that mainly relied on multilayer perceptrons or recurrent structures [2], [14]–[16].

D. Reward Design

A multi-objective reward function is designed to incorporate both physiological regulation and safety constraints into policy learning. The reward encourages the controller to maintain physiologically appropriate perfusion and stable hemodynamic behavior under changing operating conditions. At the same time, it penalizes unsafe or undesirable states associated with suction, regurgitation, and hemolysis, so that the learned policy does not improve one objective at the expense of clinical safety. This formulation extends prior DRL-based VAD reward designs that often emphasized a narrower subset of control objectives [2], [14], [16].

This reward design reflects the multi-objective nature of VAD physiological control. In practice, maintaining a single indicator within a desirable range is insufficient, because effective support must simultaneously account for perfusion demand, cardiovascular stability, and complication avoidance. Therefore, the proposed reward mechanism translates clinical and physiological considerations into learning signals that guide the controller toward safer and more adaptive behavior. By embedding these constraints into the reinforcement learning objective, the method aims to achieve a better balance between support performance and operational safety.

IV. Experimental Setup

The experimental study is designed to evaluate the proposed controller under physiologically relevant operating conditions and to examine its adaptability, robustness, and safety. Following the overall technical route of this work, the experimental setup combines a controllable simulation or mock circulatory loop environment with staged model development, including offline pretraining and online reinforcement learning.

A. Platform and Physiological Scenarios

The evaluation platform is designed to reproduce the major hemodynamic characteristics of the cardiovascular system under different physiological conditions. In this study, a controllable simulation environment and a physical mock circulatory loop are considered as complementary testbeds for controller development and validation. The simulation setting provides a flexible environment for large-scale policy interaction during reinforcement learning, whereas the mock circulatory loop offers a more realistic experimental condition with flow inertia, sensor noise, and physical disturbances. This combined setting is intended to narrow the gap between algorithm development and practical implementation.

By adjusting resistance, compliance, preload-related conditions, and other platform parameters, the experiments are configured to represent multiple operating scenarios relevant to VAD support. These scenarios include different levels of heart failure severity, changes in physiological demand associated with rest and activity, and transient operating conditions that require rapid controller adaptation. As a result, the proposed controller is exposed to a broad range of hemodynamic states rather than being validated only in a narrow steady-state regime.

As shown in Fig. 1, the physical mock circulatory loop platform includes a pulsatile motor, a volume chamber, a compliance chamber, a damping chamber, a flowmeter, and a left ventricular chamber. These components are used to emulate ventricular pumping behavior, vascular compliance, pressure buffering, and flow measurement within a controllable in-vitro environment. The platform therefore provides a practical basis for evaluating whether the proposed controller can maintain appropriate support while responding to realistic disturbances and physiological variations.

B. Data Collection and Training Procedure

The training pipeline follows a staged strategy to improve learning efficiency and stability. In the first stage, representative physiological data are collected under multiple operating conditions by varying platform parameters and recording pressure- and flow-related signals. These data provide broad

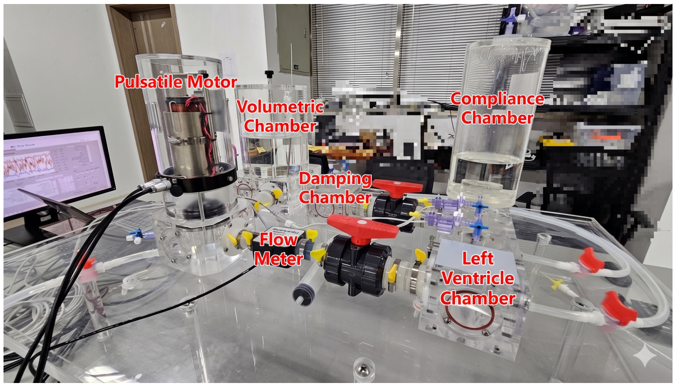

Fig. 1. Physical mock circulatory loop platform for ventricular assist device physiological control experiments. The platform includes a pulsatile motor, volume chamber, compliance chamber, damping chamber, flowmeter, and left ventricular chamber to emulate cardiovascular hemodynamic conditions.

coverage of different hemodynamic responses and are used to initialize the CNN-Transformer-based representation module before policy optimization. Such offline preparation helps the model capture informative waveform characteristics and reduces the difficulty of learning useful state representations entirely from random exploration.

In the second stage, the controller is further optimized through online reinforcement learning by interacting directly with the simulation environment and, when applicable, the physical test platform. During this process, the actor-critic policy continuously adjusts pump speed according to the observed physiological state, while the reward function guides learning toward adequate perfusion, stable hemodynamics, and reduced risk of unsafe events. This staged procedure is adopted because VAD physiological control involves highly nonlinear dynamics and multi-objective optimization, making purely end-to-end training from scratch more difficult and potentially less stable.

C. Baseline and Evaluation Protocol

The proposed method is evaluated against representative conventional and learning-based control strategies in order to examine both effectiveness and generality. Baseline methods are selected to reflect the main categories of VAD regulation approaches, including fixed-speed or rule-based controllers and existing deep reinforcement learning strategies. This comparison makes it possible to determine whether the proposed method provides meaningful improvements beyond both traditional control logic and previously reported learning-based designs.

The evaluation protocol includes three complementary parts: comparative experiments, robustness analysis, and ablation study. Comparative experiments are used to assess overall control quality under different physiological scenarios. Robustness analysis focuses on challenging conditions such as parameter perturbation, measurement noise, and abrupt physiological changes, which are important for determining whether the controller remains stable in safety-critical situations. The ablation study separately removes or replaces the CNN and Transformer components so that their individual contributions to physiological state representation and control performance can be examined.

The main evaluation criteria include physiological adaptability, hemodynamic regulation quality, control smoothness, and safety-related behavior. Particular attention is given to whether the controller can maintain appropriate support across changing operating conditions while avoiding unsafe regimes associated with suction, regurgitation, and hemolysis. Through this protocol, the experimental study is intended to provide a systematic assessment of both the effectiveness and the practical reliability of the proposed physiological control framework.

V. Results and Discussion

The experimental results are discussed from three aspects: comparative control performance, robustness and safety, and architectural ablation. This organization follows the evaluation protocol described in the previous section and is intended to provide a systematic interpretation of how the proposed controller behaves under different physiological conditions. Rather than focusing only on average performance, the discussion emphasizes whether the learned policy can maintain adaptive and stable VAD support across changing operating states.

A. Comparative Control Performance

In the comparative experiments, the proposed method is analyzed against representative baseline controllers to examine its ability to regulate pump speed under different physiological scenarios. The main point of comparison is whether the controller can better match circulatory support to the underlying hemodynamic demand during transitions such as rest-to-activity changes or different levels of cardiac impairment. From this perspective, improved performance is reflected not only by more appropriate perfusion-related outcomes, but also by the ability to preserve overall hemodynamic stability rather than optimizing a single variable in isolation.

An important observation in this context is the dynamic quality of the control response. For physiological VAD regulation, a desirable controller should respond rapidly enough to changing demand while avoiding excessive oscillation or abrupt pump speed fluctuations. Therefore, the comparative discussion focuses on both responsiveness and smoothness of adaptation. If the proposed method shows more consistent behavior across a wide range of scenarios, this would support the view that the learned policy captures meaningful physiological state information and can use that information to produce more adaptive control actions than conventional or weakly adaptive baselines.

B. Robustness and Safety Analysis

Because VAD control is inherently safety-critical, overall performance must be interpreted together with robustness under disturbances and the avoidance of unsafe operating regimes. For this reason, the results are also discussed under challenging conditions such as sensor noise, parameter variation, and abrupt physiological abnormalities. These scenarios are particularly important because a controller that performs well only under nominal conditions may still be unsuitable for practical use.

From the safety perspective, the key question is whether the proposed controller can maintain adequate support while reducing the tendency to enter regimes associated with suction, regurgitation, and hemolysis. A favorable outcome is not simply a higher level of circulatory assistance, but a better balance between support effectiveness and operational safety. If the policy exhibits more stable behavior and lower sensitivity to disturbances than the baselines, this would suggest that the multi-objective reward design successfully guides learning toward safer control strategies and that the learned policy generalizes beyond narrowly defined operating conditions.

C. Ablation Study and Interpretation

The ablation analysis is used to clarify the contribution of the CNN and Transformer modules to the overall control framework. By comparing the complete model with variants in which local feature extraction or long-range temporal modeling is weakened or removed, the experiments can reveal whether both components are necessary for accurate physiological state representation. This analysis is important because the proposed method is motivated not only by the use of reinforcement learning itself, but also by the need for a richer representation of sequential pressure and flow signals.

If the ablation results show degraded control quality after removing either component, this would indicate that local waveform morphology and cross-cycle temporal dependency both play meaningful roles in VAD physiological control. In that case, the CNN module can be interpreted as supporting the extraction of short-term hemodynamic features, whereas the Transformer module contributes to the understanding of longer-term physiological evolution. Taken together, these findings would explain why the proposed CNN-Transformer actor-critic architecture provides a more suitable basis for adaptive and safety-aware pump speed regulation than simpler network designs.

VI. Conclusion

This paper presents a deep reinforcement learning-based physiological control framework for ventricular assist devices. To overcome the limited adaptability of conventional control strategies, the proposed method integrates an actor-critic learning scheme with a CNN-Transformer state representation model and a multi-objective reward design. In this way, the framework is intended to improve pump speed regulation under nonlinear, time-varying, and safety-critical cardiovascular conditions.

The overall study establishes a systematic pipeline including physiologically relevant platform construction, staged model training, and multi-dimensional evaluation of adaptability, hemodynamic regulation quality, robustness, and safety-related behavior. Taken together, these elements provide a foundation for more adaptive and safety-aware VAD control. Future work will focus on extensive experimental validation, more complex pathological scenarios, and deployment on embedded and physical platforms for practical biomedical applications.

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